

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 08 · Straight Lines and Coordinate Geometry

With Answers

Questions with full worked solutions

- 45 pages
- Past papers up to June 2026

Straight Lines and Coordinate Geometry

Your classified route

18

questions

135

marks

4

families

11

variants

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CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Straight Lines and Coordinate Geometry

Families and variants

F01 • Linear models

V01 Fit a two-point linear model and interpret or predict

F02 • Constructing related lines

V01 Line through two points in a requested form

V02 Parallel or perpendicular line through a point

V03 Perpendicular bisector from endpoint coordinates

V04 Exact line work with surd coordinates

F03 • Coordinate intersections, shapes, distance and area

V01 Related-line intersection and perpendicular distance

V02 Coordinate polygons from vectors and diagonal midpoints

V03 Coordinate lengths and areas under geometric constraints

F04 • Straight lines with curves and regions

V01 Build line/curve boundaries and describe a region

V02 Tangent parallel to a constructed line

V03 Parameterized line-quadratic intersections with a perpendicular constraint

Linear models

WHAT THIS FAMILY ASKS

Fit and interpret $y=mx+c$ from two contextual observations.

VARIANT ROUTE

V01 Fit a two-point linear model and interpret or predict

2 Complete Classified questions

10 total marks

2. A tree was planted in the ground.

Exactly 2 years after it was planted, the height of the tree was 1.85 m.

Exactly 7 years after it was planted, the height of the tree was 3.45 m.

Given that the height, H metres, of the tree, t years after it was planted in the ground, can be modelled by the equation

$$H = at + b$$

where a and b are constants,

- (a) find the value of a and the value of b .

(4)

- (b) State, according to the model, the height of the tree when it was planted.

(1)

(Total 5 marks)

Question 1

WMA11/01 October 2019

T08 Straight Lines and Coordinate Geometry F01 Linear models

V01 Fit a two-point linear model and interpret or predict

1

Form two equations

Using $H = at + b$: when $t = 2$, $2a + b = 1.85$; when $t = 7$, $7a + b = 3.45$.

2

Solve for the gradient of the model

Subtract the first equation from the second: $5a = 1.60$, so $a = 0.32$.

3

Find the starting height

Substitute into $2a + b = 1.85$: $0.64 + b = 1.85$, hence $b = 1.21$. At planting, $t = 0$, so $H = b = 1.21$ m.

FINAL ANSWER

$a = 0.32$, $b = 1.21$. Height when planted: 1.21 m.

3. The share price of a company is monitored.

Exactly 3 years after monitoring began, the share price was £1.05

Exactly 5 years after monitoring began, the share price was £1.65

The share price, £ V , of the company is modelled by the equation

$$V = pt + q$$

where t is the number of years after monitoring began and p and q are constants.

- (a) Find the value of p and the value of q .

(3)

Exactly T years after monitoring began, the share price was £2.50

- (b) Find the value of T , according to the model, giving your answer to one decimal place.

(2)

(Total for Question 3 is 5 marks)

Question 2

WMA11/01 October 2022

T08 Straight Lines and Coordinate Geometry F01 Linear models

V01 Fit a two-point linear model and interpret or predict

1

Use the two data points

The model $V = pt + q$ passes through (3, 1.05) and (5, 1.65).

2

Find the gradient

$$p = \frac{1.65 - 1.05}{5 - 3} = \frac{0.60}{2} = 0.30.$$

3

Find the intercept

Using $1.05 = 0.30(3) + q$, we get $q = 0.15$.

4

Find T

For $V = 2.50$, $2.50 = 0.30T + 0.15$, so $T = \frac{2.35}{0.30} = 7.833 \dots$

5

Round correctly

To one decimal place, $T = 7.8$.

FINAL ANSWER

(a) $p = 0.3$, $q = 0.15$. (b) $T = 7.8$ years.

Constructing related lines

WHAT THIS FAMILY ASKS

Find line equations from points and gradients, including exact parallel, perpendicular and perpendicular-bisector constructions in the requested algebraic form.

VARIANT ROUTE

V03 Perpendicular bisector from endpoint coordinates

V04 Exact line work with surd coordinates

2 Complete Classified questions

11 total marks

1. The line l_1 passes through the point $A(-5, 20)$ and the point $B(3, -4)$.

- (a) Find an equation for l_1 giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

The line l_2 is perpendicular to l_1 and passes through the midpoint of AB

- (b) Find an equation for l_2 giving your answer in the form $px + qy + r = 0$, where p , q and r are integers.

(3)

(Total for Question 1 is 6 marks)

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Question 3

WMA11/01 October 2024

T08 Straight Lines and Coordinate Geometry

F02 Constructing lines

related

V03 Perpendicular bisector from endpoint coordinates

1

Find the gradient of l_1

$$m = \frac{-4 - 20}{3 - (-5)} = \frac{-24}{8} = -3.$$

2

Find the equation of l_1 Using $B(3, -4)$, $y = -3x + c$. Then $-4 = -9 + c$, so $c = 5$.

3

Find the midpoint of AB

$$M = \left(\frac{-5 + 3}{2}, \frac{20 + (-4)}{2} \right) = (-1, 8).$$

4

Use perpendicular gradient

Since l_1 has gradient -3 , l_2 has gradient $\frac{1}{3}$.

5

Find the equation of l_2

$$y - 8 = \frac{1}{3}(x + 1), \text{ so } 3y - 24 = x + 1 \text{ and } x - 3y + 25 = 0.$$

FINAL ANSWER

$$(a) y = -3x + 5. \quad (b) x - 3y + 25 = 0.$$

2.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Given that

- the point A has coordinates $(-2\sqrt{3}, 5)$
 - the point B has coordinates $(7\sqrt{3}, 8)$
 - the straight line l_1 passes through A and B
- (a) show that the gradient of l_1 is $p\sqrt{3}$, where p is a rational constant to be found.
You must show each step of your working.

(2)

The straight line l_2 is perpendicular to l_1 and passes through A .

- (b) Find the equation of l_2 , giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

(Total for Question 2 is 5 marks)

Question 4

WMA11/01 January 2025

T08 Straight Lines and Coordinate Geometry

F02 Constructing lines

related

V04 Exact line work with surd coordinates

1

Find the gradient of l_1

$$m = \frac{8 - 5}{7\sqrt{3} - (-2\sqrt{3})} = \frac{3}{9\sqrt{3}} = \frac{1}{3\sqrt{3}} = \frac{\sqrt{3}}{9}.$$

2

Find the perpendicular gradient

The gradient of l_2 is the negative reciprocal of $\frac{\sqrt{3}}{9}$, so $m_2 = -\frac{9}{\sqrt{3}} = -3\sqrt{3}$.

3

Use point A

l_2 passes through $A(-2\sqrt{3}, 5)$, so $y - 5 = -3\sqrt{3}(x + 2\sqrt{3})$.

4

Simplify

$y - 5 = -3\sqrt{3}x - 18$, hence $y = -3\sqrt{3}x - 13$.

FINAL ANSWER

(a) gradient = $\frac{1}{9}\sqrt{3}$. (b) $y = -3\sqrt{3}x - 13$.

Coordinate intersections, shapes, distance and area

WHAT THIS FAMILY ASKS

Use line equations together with midpoint, vector, distance and area facts to solve coordinate-geometry configurations.

VARIANT ROUTE

V01 Related-line intersection and perpendicular distance

V02 Coordinate polygons from vectors and diagonal midpoints

V03 Coordinate lengths and areas under geometric constraints

9 Complete Classified questions

71 total marks

5. The line l_1 has equation $3y - 2x = 30$

The line l_2 passes through the point $A(24, 0)$ and is perpendicular to l_1

Lines l_1 and l_2 meet at the point P

- (a) Find, using algebra and showing your working, the coordinates of P .

(5)

Given that l_1 meets the x -axis at the point B ,

- (b) find the area of triangle BPA .

(3)

(Total 8 marks)

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Question 5

WMA11/01 October 2021

T08 Straight Lines and Coordinate Geometry

F03 Coordinate intersections, shapes, distance and area

V01 Related-line intersection and perpendicular distance

1

Rewrite l_1

$3y - 2x = 30$ gives $y = \frac{2}{3}x + 10$, so the gradient of l_1 is $\frac{2}{3}$.

2

Find l_2

Since l_2 is perpendicular to l_1 , its gradient is $-\frac{3}{2}$. Through $A(24, 0)$, $y = -\frac{3}{2}(x - 24) = -\frac{3}{2}x + 36$.

3

Solve for P

At the intersection, $\frac{2}{3}x + 10 = -\frac{3}{2}x + 36$. Multiplying by 6 gives $4x + 60 = -9x + 216$, so $13x = 156$ and $x = 12$.

4

Find the y-coordinate

$y = \frac{2}{3}(12) + 10 = 18$, so $P = (12, 18)$.

5

Find B

For the x-intercept of l_1 , set $y = 0$: $-2x = 30$, so $x = -15$. Hence $B = (-15, 0)$.

6

Find the triangle area

The base BA has length $24 - (-15) = 39$, and the height from P to the x-axis is 18. Area = $\frac{1}{2}(39)(18) = 351$.

FINAL ANSWER

(a) $P = (12, 18)$. (b) 351 square units.

5.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The line l_1 has equation

$$x - 2y + 25 = 0$$

The line l_2 passes through the origin and is perpendicular to l_1

(a) Find an equation for l_2 (2)

The lines l_1 and l_2 intersect at the point P .

(b) Use algebra to find the coordinates of P . (3)

(c) Hence find the shortest distance from l_1 to the origin.

Write your answer as a fully simplified surd. (2)

(Total for Question 5 is 7 marks)

Question 6

WMA11/01 May/June 2025

T08 Straight Lines and Coordinate Geometry

F03 Coordinate intersections, shapes, distance and area

V01 Related-line intersection and perpendicular distance

1

Find the gradient of l_1 $l_1 : x - 2y + 25 = 0$ gives $y = \frac{1}{2}x + \frac{25}{2}$, so its gradient is $\frac{1}{2}$.

2

Use perpendicular gradientA line perpendicular to l_1 has gradient -2 .

3

Use the originSince l_2 passes through the origin, its equation is $y = -2x$.

4

Find the intersectionSubstitute $y = -2x$ into $x - 2y + 25 = 0$ to get $x + 4x + 25 = 0$.

5

Coordinates of I $5x + 25 = 0$, so $x = -5$ and $y = 10$. Hence $I = (-5, 10)$.

6

Shortest distance from I to originThe distance is $\sqrt{(-5)^2 + 10^2} = \sqrt{125} = 5\sqrt{5}$.

FINAL ANSWER

(a) $y = -2x$. (b) $I = (-5, 10)$. (c) $5\sqrt{5}$.

2. The line l_1 has equation $y = 10 - 3x$

The line l_2

- is perpendicular to l_1
- intersects l_1 at the point P with x coordinate 4

(a) Find an equation for l_2 writing the answer in the form $ax + by + c = 0$, where a , b and c are integers.

(3)

Given that

- line l_1 cuts the y -axis at the point M
- line l_2 cuts the y -axis at the point N

(b) find the area of triangle MPN , making your method clear.

(3)

(Total for Question 2 is 6 marks)

Question 7

WMA11/01 May/June 2026

T08 Straight Lines and Coordinate Geometry

F03 Coordinate intersections, shapes, distance and area

V01 Related-line intersection and perpendicular distance

1

(a) Locate the intersection point and perpendicular gradient

At $x = 4$ on l_1 , $y = 10 - 3(4) = -2$, so $P = (4, -2)$. The gradient of l_1 is -3 , so the gradient of the perpendicular line l_2 is $\frac{1}{3}$.

2

(a) Form the equation of the second line

$y + 2 = \frac{1}{3}(x - 4)$. Multiplying by 3 and rearranging gives $x - 3y - 10 = 0$.

3

(b) Find the two intercepts on the y-axis

For l_1 , setting $x = 0$ gives $M = (0, 10)$. From l_2 , $y = \frac{x}{3} - \frac{10}{3}$, so $N = \left(0, -\frac{10}{3}\right)$.

4

(b) Use a vertical base and horizontal height

$MN = 10 - \left(-\frac{10}{3}\right) = \frac{40}{3}$. The perpendicular distance from P to the y -axis is 4, so Area $= \frac{1}{2} \times \frac{40}{3} \times 4 = \frac{80}{3}$.

FINAL ANSWER

(a) $x - 3y - 10 = 0$. (b) $\frac{80}{3}$ square units.

6. The line l_1 has equation $3x - 4y + 20 = 0$

The line l_2 cuts the x -axis at $R(8,0)$ and is parallel to l_1

- (a) Find the equation of l_2 , writing your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(3)

The line l_1 cuts the x -axis at P and the y -axis at Q .

Given that $PQRS$ is a parallelogram, find

- (b) the area of $PQRS$,

(3)

- (c) the coordinates of S .

(2)

(Total 8 marks)

Question 8

WMA11/01 January 2020

T08 Straight Lines and Coordinate Geometry

F03 Coordinate intersections, shapes, distance and area

V02 Coordinate polygons from vectors and diagonal midpoints

1

Find the gradient of l_1 $3x - 4y + 20 = 0$ gives $y = \frac{3}{4}x + 5$, so the gradient is $\frac{3}{4}$.

2

Find l_2 through RLine l_2 is parallel to l_1 and passes through $R(8, 0)$, so $y = \frac{3}{4}(x - 8) = \frac{3}{4}x - 6$. Therefore $3x - 4y - 24 = 0$.

3

Find P and QFor l_1 , the x-intercept is $P(-\frac{20}{3}, 0)$ and the y-intercept is $Q = (0, 5)$.

4

Find the area of PQRSBase $PR = 8 - (-\frac{20}{3}) = \frac{44}{3}$ and height $OQ = 5$. Area = $\frac{44}{3} \cdot 5 = \frac{220}{3}$.

5

Find S using a vectorIn parallelogram PQRS, $\vec{PS} = \vec{QR} = (8, -5)$. Hence $S = P + (8, -5) = (-\frac{20}{3} + 8, -5) = (\frac{4}{3}, -5)$.

FINAL ANSWER

(a) $3x - 4y - 24 = 0$. (b) $\frac{220}{3}$. (c) $S = (\frac{4}{3}, -5)$.

8. The line l_1 has equation

$$2x - 5y + 7 = 0$$

(a) Find the gradient of l_1

(1)

Given that

- the point A has coordinates $(6, -2)$
- the line l_2 passes through A and is perpendicular to l_1

(b) find the equation of l_2 giving your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

The lines l_1 and l_2 intersect at the point M .

(c) Using algebra and showing all your working, find the coordinates of M .

(Solutions relying on calculator technology are not acceptable.)

(3)

Given that the diagonals of a square $ABCD$ meet at M ,

(d) find the coordinates of the point C .

(2)

(Total 9 marks)

Question 9

WMA11/01 January 2022

T08 Straight Lines and Coordinate Geometry

F03 Coordinate intersections, shapes, distance and area

V02 Coordinate polygons from vectors and diagonal midpoints

1

Find the gradient of l1 $2x - 5y + 7 = 0$ gives $y = \frac{2}{5}x + \frac{7}{5}$, so the gradient is $\frac{2}{5}$.

2

Find l2The gradient perpendicular to $\frac{2}{5}$ is $-\frac{5}{2}$. Through $A(6, -2)$, $y + 2 = -\frac{5}{2}(x - 6)$, so $y = -\frac{5}{2}x + 13$.

3

Find MAt the intersection, $\frac{2}{5}x + \frac{7}{5} = -\frac{5}{2}x + 13$. Multiplying by 10 gives $4x + 14 = -25x + 130$, so $x = 4$.

4

Find the y-coordinate of M $y = \frac{2}{5}(4) + \frac{7}{5} = 3$, so $M = (4, 3)$.

5

Use midpoint of the diagonalIn a square, the diagonals bisect each other, so M is the midpoint of AC . Thus $C = 2M - A = (8, 6) - (6, -2) = (2, 8)$.

FINAL ANSWER

(a) $\frac{2}{5}$. (b) $y = -\frac{5}{2}x + 13$. (c) $M = (4, 3)$. (d) $C = (2, 8)$.

2. The points P , Q and R have coordinates $(-3, 7)$, $(9, 11)$ and $(12, 2)$ respectively.

(a) Prove that angle $PQR = 90^\circ$ (3)

Given that the point S is such that $PQRS$ forms a rectangle,

(b) find the coordinates of S . (2)

(Total for Question 2 is 5 marks)

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Question 10

WMA11/01 January 2023

T08 Straight Lines and Coordinate Geometry

F03 Coordinate intersections, shapes, distance and area

V02 Coordinate polygons from vectors and diagonal midpoints

1

Find the gradient of PQ

$$m_{PQ} = \frac{11 - 7}{9 - (-3)} = \frac{4}{12} = \frac{1}{3}.$$

2

Find the gradient of QR

$$m_{QR} = \frac{2 - 11}{12 - 9} = \frac{-9}{3} = -3.$$

3

Prove the right angle

$$m_{PQ}m_{QR} = \frac{1}{3}(-3) = -1, \text{ so } PQ \text{ and } QR \text{ are perpendicular. Hence angle } PQR = 90^\circ.$$

4

Use the rectangle vector

$$\text{For rectangle } PQRS, \overrightarrow{PS} = \overrightarrow{QR} = (12 - 9, 2 - 11) = (3, -9).$$

5

Find S

$$S = P + \overrightarrow{QR} = (-3, 7) + (3, -9) = (0, -2).$$

FINAL ANSWER

(a) Angle $PQR = 90^\circ$. (b) $S = (0, -2)$.

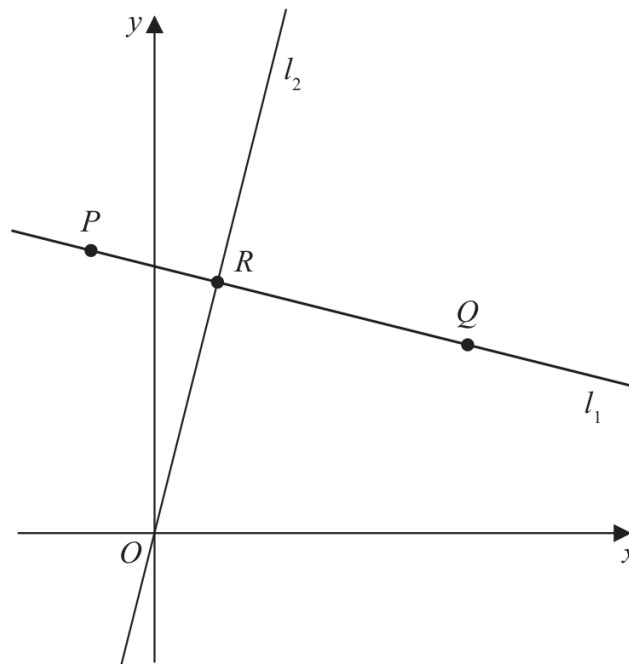


Figure 1

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The straight line l_1 , shown in Figure 1, passes through the points $P(-2, 9)$ and $Q(10, 6)$.

- (a) Find the equation of l_1 , giving your answer in the form $y = mx + c$, where m and c are constants to be found. (3)

The straight line l_2 passes through the origin O and is perpendicular to l_1

The lines l_1 and l_2 meet at the point R as shown in Figure 1.

- (b) Find the coordinates of R (4)

- (c) Find the exact area of triangle OPQ . (3)

(Total for Question 5 is 10 marks)

Question 11

WMA11/01 January 2024

T08 Straight Lines and Coordinate Geometry

F03 Coordinate intersections, shapes, distance and area

V03 Coordinate lengths and areas under geometric constraints

1

Find the gradient of l_1

$$m = \frac{6 - 9}{10 - (-2)} = -\frac{3}{12} = -\frac{1}{4}.$$

2

Find the equation of l_1

Using $P(-2, 9)$, $y = -\frac{1}{4}x + c$. Then $9 = \frac{1}{2} + c$, so $c = \frac{17}{2}$.

3

Find the equation of l_2

A line perpendicular to gradient $-\frac{1}{4}$ has gradient 4. Since l_2 passes through the origin, l_2 is $y = 4x$.

4

Find R

At the intersection, $4x = -\frac{1}{4}x + \frac{17}{2}$. Hence $\frac{17}{4}x = \frac{17}{2}$, so $x = 2$ and $y = 8$.

5

Use coordinate area

For $O(0, 0)$, $P(-2, 9)$ and $Q(10, 6)$, the area is $\frac{1}{2}|(-2)(6) - 9(10)| = 51$.

FINAL ANSWER

(a) $y = -\frac{1}{4}x + \frac{17}{2}$. (b) $R = (2, 8)$. (c) **Area** = 51.

9. Given that

- the point A has coordinates $(4, 2)$
- the point B has coordinates $(15, 7)$
- the line l_1 passes through A and B

(a) find an equation for l_1 , giving your answer in the form $px + qy + r = 0$ where p , q and r are integers to be found.

(3)

The line l_2 passes through A and is parallel to the x -axis.

The point C lies on l_2 so that the length of BC is $5\sqrt{5}$

(b) Find both possible pairs of coordinates of the point C .

(4)

(c) Hence find the minimum possible area of triangle ABC .

(2)

(Total for Question 9 is 9 marks)

Question 12

WMA11/01 May/June 2024

T08 Straight Lines and Coordinate Geometry

F03 Coordinate intersections, shapes, distance and area

V03 Coordinate lengths and areas under geometric constraints

1

Find the gradient of l_1

$$m = \frac{7-2}{15-4} = \frac{5}{11}.$$

2

Use point A

$$y - 2 = \frac{5}{11}(x - 4).$$

3

Rearrange

$$11y - 22 = 5x - 20, \text{ so } 5x - 11y + 2 = 0.$$

4

Use the horizontal line l_2

Since l_2 passes through $A(4, 2)$ and is parallel to the x-axis, point C has coordinates $(c, 2)$.

5

Use BC equals 5 root 5

$$(c - 15)^2 + (2 - 7)^2 = (5\sqrt{5})^2.$$

6

Solve for c

$$(c - 15)^2 + 25 = 125, \text{ so } (c - 15)^2 = 100 \text{ and } c = 5 \text{ or } 25.$$

7

Find the smaller area

The height from B to $y = 2$ is 5. For $C = (5, 2)$, $AC = 1$, so the minimum area is $\frac{1}{2}(1)(5) = \frac{5}{2}$.

FINAL ANSWER

(a) $5x - 11y + 2 = 0$. (b) $C = (5, 2)$ **or** $(25, 2)$. (c) $\frac{5}{2}$.

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4.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The point $A(-1, 8)$ and the point $B(7, 2)$ lie on the line L_1

- (a) Find an equation for L_1 in the form $ax + by + c = 0$, where a , b and c are integers. (3)

The line L_2

- passes through A and is perpendicular to L_1
- cuts the x -axis at the point C

- (b) Find the coordinates of C . (3)

- (c) Hence find the area of triangle ABC . (3)

(Total for Question 4 is 9 marks)

Question 13

WMA11/01 January 2026

T08 Straight Lines and Coordinate Geometry

F03 Coordinate intersections, shapes, distance and area

V03 Coordinate lengths and areas under geometric constraints

1

Gradient of L1

Using $A(-1, 8)$ and $B(7, 2)$, $m = \frac{2 - 8}{7 - (-1)} = -\frac{6}{8} = -\frac{3}{4}$.

2

Equation of L1

$$y - 2 = -\frac{3}{4}(x - 7).$$

3

Rearrange

$$4y - 8 = -3x + 21, \text{ hence } 3x + 4y - 29 = 0.$$

4

Gradient of L2

Since L_2 is perpendicular to L_1 , its gradient is $\frac{4}{3}$.

5

Equation of L2

$$L_2 \text{ passes through } A(-1, 8), \text{ so } y - 8 = \frac{4}{3}(x + 1).$$

6

Find C

At the x-axis, $y = 0$, so $-8 = \frac{4}{3}(x + 1)$. Hence $x + 1 = -6$ and $x = -7$, giving $C = (-7, 0)$.

7

Find perpendicular side lengths

$$AB = \sqrt{8^2 + (-6)^2} = 10 \text{ and } AC = \sqrt{(-6)^2 + (-8)^2} = 10.$$

8

Area of triangle ABC

Since $AB \perp AC$, **Area** = $\frac{1}{2}(10)(10) = 50$.

FINAL ANSWER

(a) $3x + 4y - 29 = 0$. (b) $C = (-7, 0)$. (c) 50.

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Straight lines with curves and regions

WHAT THIS FAMILY ASKS

Combine line construction with quadratic equations, tangent gradients or graph inequalities.

VARIANT ROUTE

V01 Build line/curve boundaries and describe a region

V02 Tangent parallel to a constructed line

V03 Parameterized line-quadratic intersections with a perpendicular constraint

5 Complete Classified questions

43 total marks

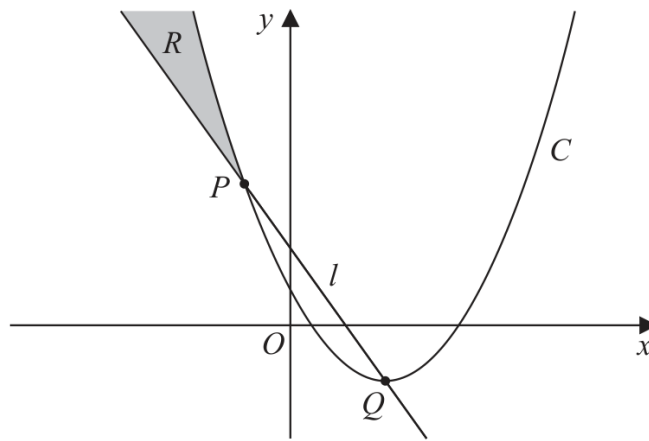


Figure 2

The points P and Q , as shown in Figure 2, have coordinates $(-2, 13)$ and $(4, -5)$ respectively.

The straight line l passes through P and Q .

- (a) Find an equation for l , writing your answer in the form $y = mx + c$, where m and c are integers to be found. (3)

The quadratic curve C passes through P and has a minimum point at Q .

- (b) Find an equation for C . (3)

The region R , shown shaded in Figure 2, lies in the second quadrant and is bounded by C and l only.

- (c) Use inequalities to define region R . (2)

(Total 8 marks)

Question 14

WMA11/01 January 2021

T08 Straight Lines and Coordinate Geometry

F04 Straight lines with curves and regions

V01 Build line/curve boundaries and describe a region

1

Find the line through P and Q

The gradient is $\frac{-5 - 13}{4 - (-2)} = \frac{-18}{6} = -3$. Using $P(-2, 13)$, $y - 13 = -3(x + 2)$, hence $y = -3x + 7$.

2

Use vertex form for the quadratic

The minimum point is $Q(4, -5)$, so the curve has form $y = a(x - 4)^2 - 5$.

3

Substitute P to find a

Since $P(-2, 13)$ lies on the curve, $13 = a(-6)^2 - 5$. Therefore $18 = 36a$ and $a = \frac{1}{2}$.

4

Write the curve

C has equation $y = \frac{1}{2}(x - 4)^2 - 5$.

5

Define the shaded region

In the shaded second-quadrant region, points are above the line, below the curve, and to the left of P. Therefore $y > -3x + 7$, $y < \frac{1}{2}(x - 4)^2 - 5$, and $x < -2$.

FINAL ANSWER

$$(a) y = -3x + 7. \quad (b) y = \frac{1}{2}(x - 4)^2 - 5. \quad (c) y > -3x + 7, y < \frac{1}{2}(x - 4)^2 - 5, x < -2.$$

3. (i) Solve

$$\frac{3}{x} > 4$$

(3)

(ii)

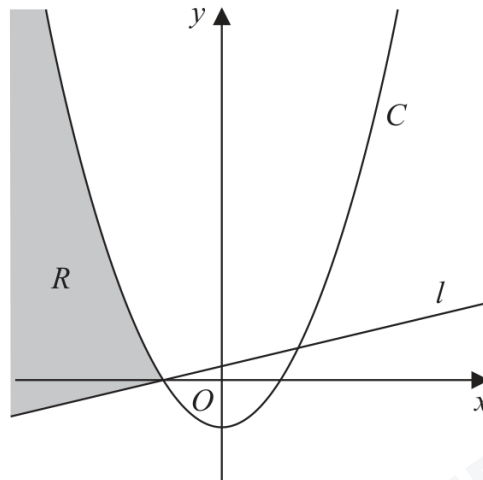


Figure 1

Figure 1 shows a sketch of the curve C and the straight line l .

The infinite region R , shown shaded in Figure 1, lies in quadrants 2 and 3 and is bounded by C and l only.

Given that

- l has a gradient of 3
- C has equation $y = 2x^2 - 50$
- C and l intersect on the negative x -axis

use inequalities to define the region R .

(3)

(Total 6 marks)

Question 15

WMA11/01 October 2021

T08 Straight Lines and Coordinate Geometry

F04 Straight lines with curves and regions

V01 Build line/curve boundaries and describe a region

1

Solve the rational inequality $\frac{3}{x} > 4$ is equivalent to $\frac{3-4x}{x} > 0$.

2

Use critical valuesThe critical values are $x = 0$ and $x = \frac{3}{4}$. A sign check gives the positive interval $0 < x < \frac{3}{4}$.

3

Find the equation of l The curve C meets the negative x -axis when $2x^2 - 50 = 0$, so $x = -5$. Since l has gradient 3 and passes through $(-5, 0)$, $y = 3(x + 5) = 3x + 15$.

4

Define the shaded regionThe shaded region lies above the line, below the curve, and to the left of their negative x -axis intersection. Hence $y > 3x + 15$, $y < 2x^2 - 50$, and $x < -5$.**FINAL ANSWER**(i) $0 < x < \frac{3}{4}$. (ii) $y > 3x + 15$, $y < 2x^2 - 50$, $x < -5$.

5. The curve C has equation $y = f(x)$

Given that

- $f(x)$ is a quadratic expression
- the maximum turning point on C has coordinates $(-2, 12)$
- C cuts the negative x -axis at -5

- (a) find $f(x)$

(4)

The line l_1 has equation $y = \frac{4}{5}x$

Given that the line l_2 is perpendicular to l_1 and passes through $(-5, 0)$

- (b) find an equation for l_2 , writing your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

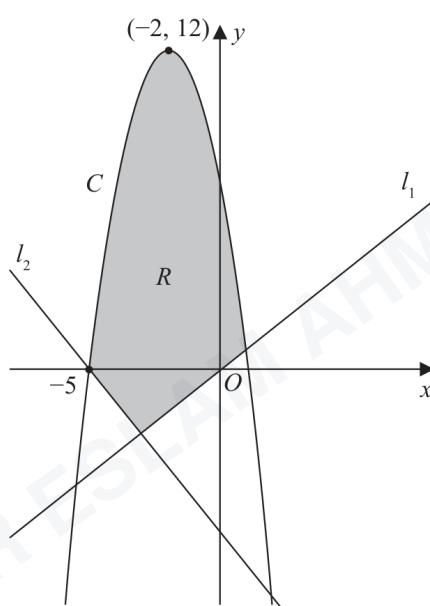


Figure 2

Figure 2 shows a sketch of the curve C and the lines l_1 and l_2

- (c) Define the region R , shown shaded in Figure 2, using inequalities.

(2)

(Total 9 marks)

Question 16

WMA11/01 May/June 2022

T08 Straight Lines and Coordinate Geometry

F04 Straight lines with curves and regions

V01 Build line/curve boundaries and describe a region

1

Use symmetry of the quadratic

The maximum is at $x = -2$, and one root is $x = -5$. The other root is the same distance to the right of -2 , so it is $x = 1$.

2

Find the quadratic

Let $f(x) = a(x + 5)(x - 1)$. Since the maximum point is $(-2, 12)$, $12 = a(3)(-3) = -9a$, so $a = -\frac{4}{3}$.

3

Find the perpendicular gradient

The gradient of l_1 is $\frac{4}{5}$, so the gradient of a perpendicular line is $-\frac{5}{4}$.

4

Find l_2

Using the point $(-5, 0)$, $y = -\frac{5}{4}x + c$. Then $0 = \frac{25}{4} + c$, so $c = -\frac{25}{4}$ and l_2 is $y = -\frac{5}{4}x - \frac{25}{4}$.

5

Write the inequalities for R

The region is below the curve and above both straight lines, so $y \leq f(x)$, $y \geq \frac{4}{5}x$, and $y \geq -\frac{5}{4}x - \frac{25}{4}$.

FINAL ANSWER

$$(a) f(x) = -\frac{4}{3}(x + 5)(x - 1). \quad (b) y = -\frac{5}{4}x - \frac{25}{4}. \quad (c) y \leq f(x), y \geq \frac{4}{5}x, y \geq -\frac{5}{4}x - \frac{25}{4}.$$

8.

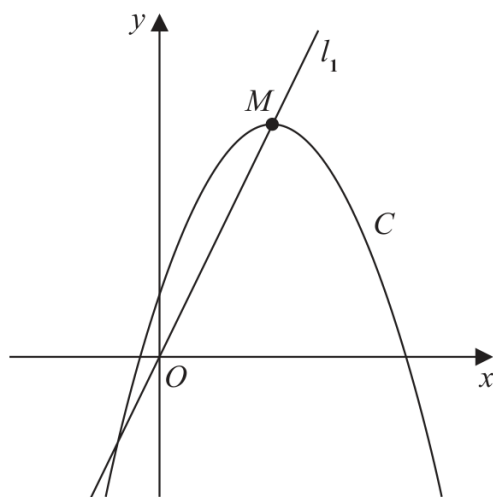


Figure 4

Figure 4 shows a sketch of the curve C with equation

$$y = 4 + 12x - 3x^2$$

The point M is the maximum turning point on C .

- (a) (i) Write $4 + 12x - 3x^2$ in the form

$$a + b(x + c)^2$$

where a , b and c are constants to be found.

- (ii) Hence, or otherwise, state the coordinates of M .

(5)

The line l_1 passes through O and M , as shown in Figure 4.

A line l_2 touches C and is parallel to l_1

- (b) Find an equation for l_2

(5)

(Total 10 marks)

Question 17

WMA11/01 October 2021

T08 Straight Lines and Coordinate Geometry

F04 Straight lines with curves and regions

V02 Tangent parallel to a constructed line

1

Complete the square

$$4 + 12x - 3x^2 = -3(x^2 - 4x) + 4 = -3((x - 2)^2 - 4) + 4 = 16 - 3(x - 2)^2.$$

2

State the maximum pointThe maximum occurs when $(x - 2)^2 = 0$, so $M = (2, 16)$.

3

Find the gradient of l_1 Line l_1 passes through $O(0, 0)$ and $M(2, 16)$, so its gradient is $\frac{16}{2} = 8$.

4

Use the parallel tangent gradientLine l_2 is parallel to l_1 , so its gradient is also 8. For $y = 4 + 12x - 3x^2$, $\frac{dy}{dx} = 12 - 6x$.

5

Find the point of contactSet $12 - 6x = 8$, giving $x = \frac{2}{3}$. Then $y = 4 + 12\left(\frac{2}{3}\right) - 3\left(\frac{4}{9}\right) = \frac{32}{3}$.

6

Write l_2 Using gradient 8 through $\left(\frac{2}{3}, \frac{32}{3}\right)$, $y - \frac{32}{3} = 8\left(x - \frac{2}{3}\right)$. Hence $y = 8x + \frac{16}{3}$.

FINAL ANSWER

(a)(i) $16 - 3(x - 2)^2$. (a)(ii) $M = (2, 16)$. (b) $y = 8x + \frac{16}{3}$.

10.

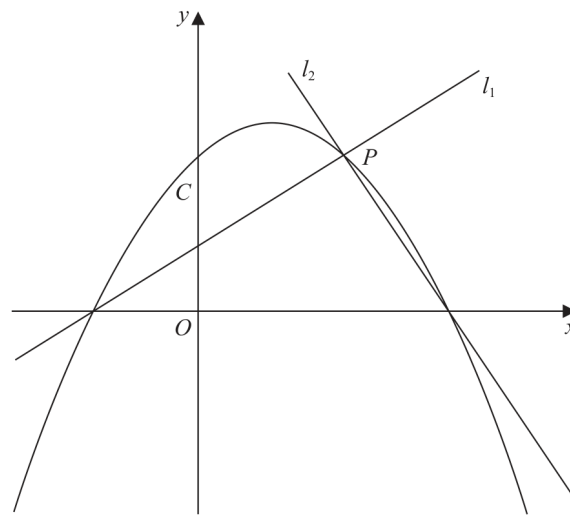


Figure 5

Figure 5 shows a sketch of the quadratic curve C with equation

$$y = -\frac{1}{4}(x+2)(x-b) \quad \text{where } b \text{ is a positive constant}$$

The line l_1 also shown in Figure 5,

- has gradient $\frac{1}{2}$
- intersects C on the negative x -axis and at the point P

(a) (i) Write down an equation for l_1

(1)

(ii) Find, in terms of b , the coordinates of P

(3)

Given that the line l_2 is perpendicular to l_1 and intersects C on the positive x -axis,

(b) find, in terms of b , an equation for l_2

(2)

Given also that l_2 intersects C at the point P

(c) show that another equation for l_2 is

$$y = -2x + \frac{5b}{2} - 4$$

(2)

(d) Hence, or otherwise, find the value of b

(2)

(Total for Question 10 is 10 marks)

Question 18

WMA11/01 May/June 2023

T08 Straight Lines and Coordinate Geometry

F04 Straight lines with curves and regions

V03 Parameterized line-quadratic intersections with a perpendicular constraint

1

Find l_1

Line l_1 has gradient $\frac{1}{2}$ and passes through the negative x-axis root $(-2, 0)$, so $y = \frac{1}{2}(x + 2) = \frac{1}{2}x + 1$.

2

Find P

Set $-\frac{1}{4}(x + 2)(x - b) = \frac{1}{2}(x + 2)$.

3

Solve the intersection

Multiplying by -4 gives $(x + 2)(x - b) = -2(x + 2)$, so $(x + 2)(x - b + 2) = 0$.

4

Choose the second point

The root $x = -2$ is the x-axis point, so P has $x = b - 2$. Then $y = \frac{1}{2}(b - 2) + 1 = \frac{b}{2}$.

5

Find l_2 from the positive root

Since l_2 is perpendicular to l_1 , its gradient is -2 . It passes through the positive x-axis root $(b, 0)$, so $y = -2(x - b) = -2x + 2b$.

6

Use P to write another l_2

Through $P\left(b - 2, \frac{b}{2}\right)$ with gradient -2 : $y - \frac{b}{2} = -2(x - (b - 2))$, so $y = -2x + \frac{5b}{2} - 4$.

7

Find b

Both equations represent l_2 , so $2b = \frac{5b}{2} - 4$. Hence $b = 8$.

FINAL ANSWER

$$(a)(i) y = \frac{1}{2}x + 1. \quad (a)(ii) P = \left(b - 2, \frac{b}{2}\right). \quad (b) y = -2x + 2b.$$
$$(d) b = 8.$$

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